North Country National Scenic Trail Route Plan

North Star Consulting

December 8, 2025

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I. Introduction

The North Country National Scenic Trail (NCT) is the longest continuous hiking trail in the United States, extending more than 4,800 miles across eight states and connecting a remarkable diversity of ecological, cultural, and recreational landscapes. In Minnesota, roughly 859 miles of the route traverse the northern half of the state, incorporating established segments such as the Superior Hiking Trail and the Border Route Trail. Despite this progress, significant gaps remain where the trail has not yet been constructed as a permanent, natural-surface route. One of the highest priorities is the segment between Grand Rapids and Snowbank Lake, also referred to as the Arrowhead region, where approximately 150 miles of the current alignment rely heavily on road walks and temporary connections rather than purpose-built hiking trails.

In the Arrowhead region, the present state of the trail is characterized by long stretches along state and county highways and local roads. These segments diminish scenic quality, increase exposure to vehicle traffic, and fragment the sense of immersion that long-distance hikers expect from a National Scenic Trail. At present, there is an active, long-running volunteer effort to identify and secure off-road reroutes. Volunteers, chapter leaders, and partner agencies have sketched preliminary route ideas, walked candidate corridors, and discussed possibilities with land managers, but they have had limited access to a unified geospatial framework that allows them to evaluate options across the entire corridor in a consistent way.

Through the University of Minnesota's GIS 8501 course, 18 Master of Geographic Information Science (MGIS) students formed North Star Consulting to support the North Country Trail Association (NCTA) in advancing this work. Over the course of a semester, the team developed a geospatial workflow, suitability model, and suite of map products designed to highlight where the landscape is most favorable for an off-road NCT alignment and where major constraints remain. The analysis emphasizes features that are desirable for trail development such as public land, existing non-motorized trails, campgrounds, and bridge crossings, while identifying features that are difficult or undesirable to cross, including private parcels, wetlands, large water bodies, major roads, and portions of multi-use trails that emphasize motorized or paved use.

This report summarizes that work, describing the study area and objectives, explaining how the suitability model and least-cost path analysis were constructed, and presenting results for three focus areas: Grand Rapids, the central Arrowhead region around the Taconite State Trail, and the Ely area near Snowbank Lake. The report concludes with a discussion of tools delivered to NCTA, limitations of the model, and recommendations for future ground-truthing, partnership development, and higher-resolution re-modeling. This report intended to complement the project StoryMap and web maps and to serve as a baseline document for future phases of route planning and environmental review.

II. Project Objectives and Study Area

At the beginning of the course,the class identified a shared set of objectives for the North Country Trail project. The first objective was to compile a standardized collection of environmental, infrastructural, and ownership datasets for the Grand Rapids–Snowbank Lake corridor, drawing on Minnesota Geospatial Commons, MnDOT resources, and data shared by such as the North Country Trail Association and state agencies. Second, the project aimed to construct a suitability model that classifies the landscape into more and less appropriate locations for a natural-surface hiking trail, based on factors such as public versus private ownership, proximity to existing trails, and the presence of hydrologic barriers. Third, the team aimed to use that suitability surface to generate candidate trail alignments through least-cost path analysis between key anchor points. Finally, the project sought to deliver web and desktop tools (shapefiles, KMLs, web maps, and summary tables) that NCTA and local volunteers can use in ongoing route evaluation, ground-truthing, and environmental documentation.

The study area is defined by the NCT's revised preferred alternative corridor between Grand Rapids and Snowbank Lake. This corridor polygon, provided by NCTA, represents a broad swath of land north of the existing trail within which a future permanent route is desired (Figure 1). To account for edge effects and provide routing flexibility around the corridor boundary, North Star Consulting worked within a buffered version of this corridor that extends modestly beyond the preferred alignment. The primary counties intersecting this area of interest are Itasca, St. Louis, and Lake Counties, with a small overlap into Aitkin County.

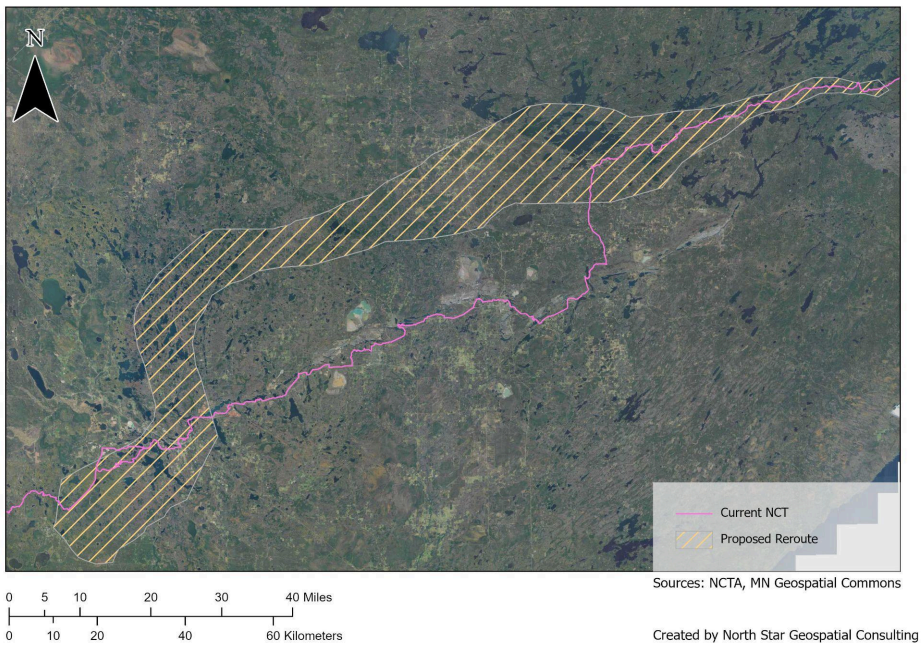


Figure 1. Location of the North Country National Scenic Trail in Minnesota and the focus segment between Grand Rapids and Snowbank Lake (pink) and the new preferred route corridor (yellow).

To keep the analysis focused and computationally efficient, parcels and other land-based datasets were restricted to the counties that clearly intersect the main routing direction between Grand Rapids and Snowbank Lake. Parcel data from opt-in open data counties and an additional Itasca parcel dataset were integrated to provide the best available coverage. Because Aitkin County's overlap with the corridor is minimal and lies south of the primary east–west routing direction, its parcels were not included in the final suitability model.

Within this corridor, the analysis focuses particularly on three subregions: Grand Rapids, the central Arrowhead region around the Taconite State Trail, and the Ely area near Snowbank Lake. Near Grand Rapids, the trail already uses a mix of off-road segments and road walks as it approaches and leaves the city. In the central Arrowhead, the current alignment relies heavily on roads but runs near the Taconite State Trail and other public lands that offer opportunities for rerouting. Near Ely and Snowbank Lake, the corridor is constrained by a combination of private land patterns and transportation infrastructure, making partnership with existing right-of-way (ROW) managers especially important.

III. Methods

In order to identify candidate NCT alignments within the corridor, North Star Consulting developed a raster-based suitability model and least-cost path workflow. The analytical framework is based on two main steps. First, diverse datasets representing ownership, recreation infrastructure, and environmental constraints were translated into a common ordinal suitability scale. Second, these datasets were combined into a composite cost surface that guides least-cost path analysis between chosen locations on the trail.

The suitability model uses a seven-class scale, where low values represent highly suitable locations, values in the middle of the scale represent neutral or moderately suitable areas, and higher values represent increasingly unsuitable or effectively excluded locations. For example, existing non-motorized hiking trails, state forest campgrounds, and public lands were generally assigned lower values, while private parcels, wetlands, lakes, and road surfaces were assigned higher values.. For each input layer, the team created a consistent classification field and applied reclassification rules that reflected the expected feasibility and desirability of routing the trail through those features (Figure 2).

Data acquisition and preparation followed a consistent pattern. First, statewide or regional datasets were gathered from Minnesota Geospatial Commons, MnDOT, and other open sources. These included parcel layers from opt-in counties, a statewide trails layer, a state forest campground dataset, a hydrology dataset containing rivers, lakes, and wetlands, and a MnDOT bridge inventory, among others (Appendix A). All layers were reprojected to a common coordinate system (NAD 1983 UTM Zone 15N) to ensure alignment during overlay and raster conversion. Each dataset was then clipped to the buffered NCT corridor or to the intersecting

counties so that only features relevant to the study area remained. This reduced processing time and avoided including features far outside the corridor that would not reasonably influence the final alignment.

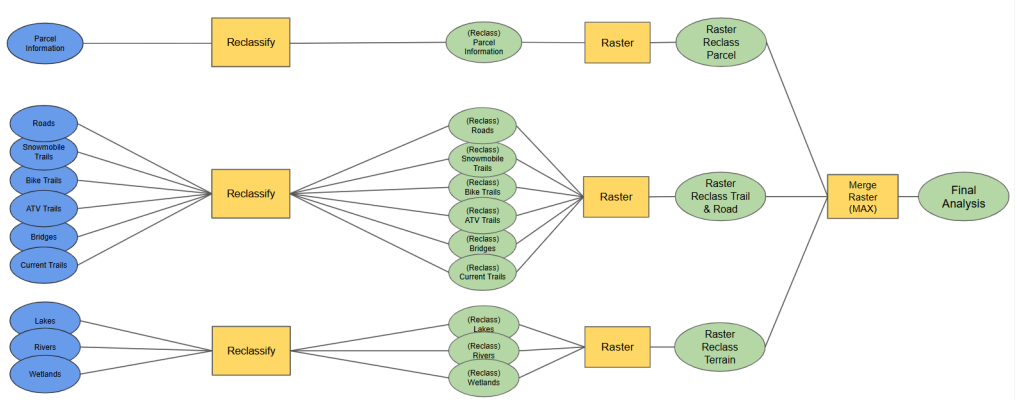


Figure 2. Data flow diagram data and methods used to create the suitability map and trail route model.

The parcel dataset required particular pre-processing. Parcels from open-data counties were combined with Itasca County parcels obtained from another source. Ownership codes were generalized into a smaller number of categories reflecting their likely feasibility for trail development. Lands held by federal, state, county, or similar public entities were grouped into a "public" category that received favorable suitability scores. Regional or special-purpose lands, such as local authorities or institutional holdings, were grouped into a more neutral category. Private parcels were placed in a category that received relatively unfavorable scores, reflecting the negotiation and uncertainty inherent in securing access across them. Aitkin County parcels were not brought into the final classification, in order to maintain focus on the main alignment between Grand Rapids and Snowbank Lake rather than encouraging large southerly detours.

Existing hiking trails were processed to highlight non-motorized hiking infrastructure that could potentially be shared with the NCT. Trails coded as allowing hiking were selected and clipped to the corridor. To allow for slight deviations around the mapped alignments, the trails were buffered by a small distance before conversion to raster. Both the trail polylines and their buffers were translated into cells with low suitability values that encourage the model to use these corridors wherever possible.

Campgrounds were processed as attractive nodes along the potential route. State forest campgrounds and day-use sites were reviewed, and sites used exclusively for horses or day use were removed. Remaining campgrounds were clipped to the corridor, buffered to represent the broader area within walking distance of each site, and assigned favorable suitability values that encourage routes to favor those areas. These buffered polygons were converted to raster and incorporated into the composite cost surface.

Bridges were treated as key crossing opportunities over rivers and highways. The MnDOT bridge inventory was filtered to retain only pedestrian and mixed-use structures, then clipped to the corridor, then buffered to create a small zone within which the model could approach the crossing. Buffers were ranked based on whether the structure primarily served pedestrians or a mix of highway and pedestrian use, and whether it crossed a water feature. This ranking gave the highest preference to purely non-motorized water crossings, while still recognizing the value of shared-use bridges that provide safe, established crossings. The buffers were converted to raster and added to the cost surface as favorable cells around viable crossing locations.

Hydrologic features were treated largely as constraints. Rivers, lakes, and wetlands were selected where they intersect the corridor then converted into a raster with high suitability scores that discourage routing through them, except where bridge buffers provide designated crossings. This approach reduces the likelihood that a modeled route will propose crossing open water or saturated ground at locations where no practical bridge or boardwalk is present.

After all layers were prepared and reclassified, they were converted to a common 30-meter raster resolution and combined into a composite cost surface using raster algebra. This step effectively stacked the influences of ownership, hydrology, trails, campgrounds, and bridges. Areas where multiple favorable features overlapped, for example public lands near an existing trail and a campground, received low combined cost values, while areas with overlapping constraints, such as private parcels covered by wetlands, received high cost values. The resulting surface describes, cell by cell, the relative “effort” required to construct and manage a trail.

Least-cost path analysis was then used to derive candidate NCT alignments. For this project, anchor points were placed where the current NCT changes from road walks to more suitable natural-surface segments and at key locations where major route decisions must be made along the corridor. Using the composite cost surface, the least-cost path algorithm identified routes that minimize cumulative cost while remaining within the corridor. These raster paths were converted back to vector polylines, visually reviewed, and compared with the current road-based alignment and with preliminary routes that volunteers had previously drafted.

Although the model provides powerful screening capabilities, it has important limitations. The choice of 30-meter resolution balances regional coverage with computation time but cannot capture fine-scale microtopography, small streams, or short steep slopes that may matter to field crews. The seven-class suitability scale compresses complex factors into a limited set of categories; for example, not all private land is equally difficult to negotiate, and not all public land is equally suitable for trail development. The model also relies on the best available open data at a single point in time, and it does not include confidential environmental or cultural resource datasets that would need to be considered in formal environmental review. These limitations reinforce that the modeled route is a tool for screening and prioritization rather than a final decision.

IV. Results

The modeling results can be understood at two scales. At the corridor scale, the suitability surface distinguishes broad trends of opportunity from broad trends of constraint, helping planners see where the landscape naturally supports an off-road NCT alignment (Figure 3). At the local scale, least-cost paths show how a candidate route might thread through these bands as it passes through specific communities and land ownership patterns.

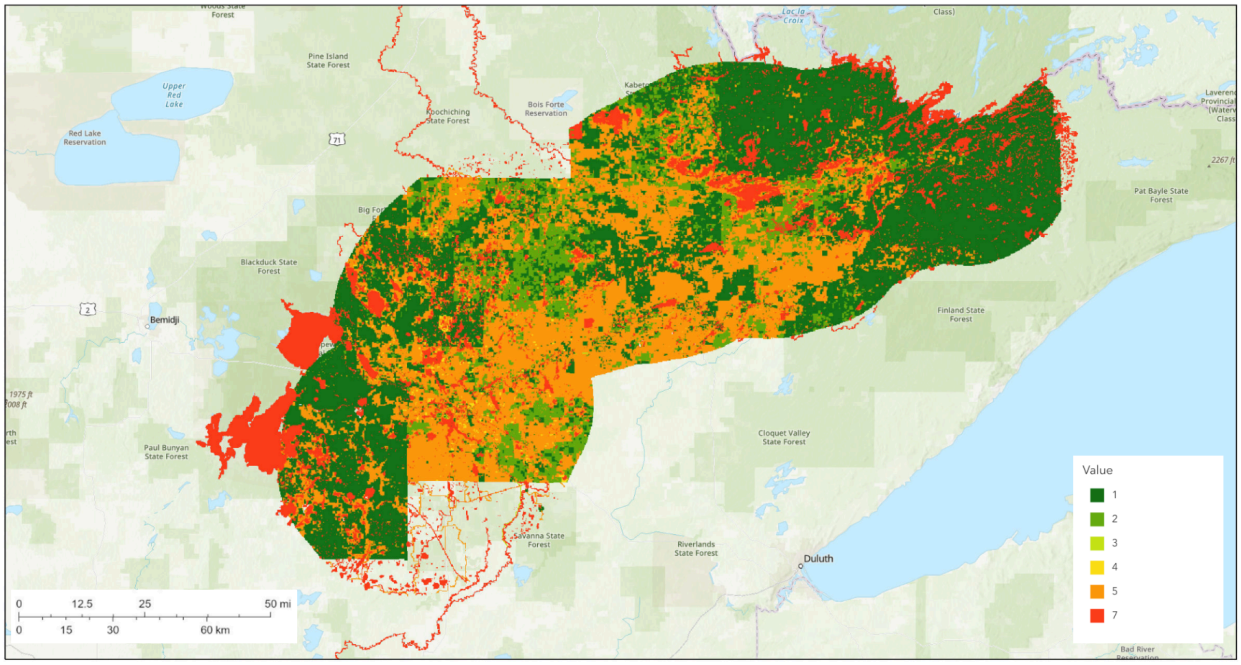


Figure 3. Final suitability raster layer classifying the route suitability of the North Country Trail Corridor where green is most suitable and red is least suitable.

In the Grand Rapids area, the current NCT alignment is relatively strong compared to other segments. Much of the trail already uses off-road corridors and passes through public land and negotiates urban traversability constraints. The suitability model reinforces this pattern: large swaths of favorable cells appear around the existing off-road segments, and the least-cost path tends to follow these areas closely. The road-based segments near the city limits and within Grand Rapids itself, however, stand out as stretches where the cost surface becomes less favorable because of denser development, more complex ownership, and a higher density of paved surfaces. For the city itself, we did not re-model detailed street-by-street options as, given present constraints, road-based travel through Grand Rapids remains the most practical option. The surrounding layers still help identify where future easements or short connecting paths might allow the trail to reach quieter streets or improved crossings in the long term, and these are decisions that local planners and chapter leaders can evaluate using their on-the-ground knowledge.

In the central Arrowhead region, where the current alignment relies heavily on roads, the suitability model reveals a different picture. The corridor provided by NCT already directed attention to the area north of the existing road-based route, and the model further supports this emphasis by highlighting large areas of public and industrial forest land, along with the presence of the Taconite State Trail. Within this band of higher suitability, the least-cost path aligns closely with preliminary volunteer-drafted routes that have been sketched as conceptual reroutes. This convergence between modeled and volunteer routes is encouraging, because it suggests that the statewide datasets and suitability rules are capturing many of the same advantages and constraints that volunteers have observed on the ground.

The Taconite State Trail itself emerged as a key feature in our analysis. The model recognizes it as an existing corridor through public land and thus assigns favorable cost values to cells in its vicinity. However, the trail's mixed-use character complicates its suitability for the NCT. All segments allow horses and bicycles, some allow all-terrain vehicles, and small sections are paved. These attributes may not fully align with the desired characteristics of a primarily pedestrian, natural-surface National Scenic Trail. At the same time, most of the Taconite corridor is natural terrain or gravel, and it already has a right-of-way and management plan in place, which can make co-location or parallel routing more feasible than securing entirely new easements. The model does not encode these policy and experience nuances in full; instead, it highlights where co-location might be geographically efficient, leaving detailed decisions about how closely and for how long to share the corridor to later planning stages.

In the Ely and Snowbank Lake area, the modeling results underscore just how constrained the corridor becomes. The composite suitability surface reveals bottlenecks where highway crossings, hydrologic barriers, and private parcels converge. Bridge and hydrology layers identify the crossing that can safely be used by hikers, and this location effectively functions as a gate through which any continuous route must pass. Beyond that crossing, the candidate route must still contend with dense private ownership and the need to weave between lakes, wetlands, and existing infrastructure. This pattern makes it clear that purely data-driven routing has limits in such settings and that strong partnerships will be as important as the suitability surface itself. There is already an existing mowed utility clearing for electrical transmission that runs through much of this area, and that corridor may offer a tangible starting point for discussions about co-located or adjacent trail segments

Because of these constraints and the relative coarseness of the available data, the Ely segment of the modeled route should be viewed as a first pass. It provides a helpful indication of where the landscape and infrastructure make a continuous off-road route most feasible, but it will need to be revisited with higher resolution terrain data, more detailed ownership information, and close collaboration with agencies and landowners before any route is advanced toward implementation.

V. Tools and Data Products

North Star Consulting delivered a set of geospatial tools meant to integrate smoothly into NCTA's existing workflow. The most direct product is a shapefile representing the modeled NCT alignment between Grand Rapids and Snowbank Lake. This file can be loaded into ArcGIS Pro, ArcGIS Online, or other GIS software for visualization and editing. A KML version of the same line was also created for easy use with GPS devices and mobile apps, allowing volunteers to download the modeled route, walk it, and record suggested edits back with project coordinators.

The project also produced a trail suitability map derived from the composite cost surface. In web map and static form, this heat map uses a progression from softer, lighter tones for less suitable areas to deeper, more saturated tones for highly suitable areas. When combined with parcel layers and point of interest locations, the suitability map becomes a useful tool for designing field visits and prioritizing which segments of the corridor warrant more detailed investigation.

A third major product is a notification-style web application built around parcels intersected by the potential trail corridor and modeled route. This web app allows planners and volunteers to see which landowners would potentially be affected by a given alignment and to begin organizing outreach and negotiation strategies. As NCTA and partners refine the route, they can update this impacted-parcels dataset to reflect new trail alignments. Additionally, the app is designed to draw on data provided by the state so that parcel-related information stays current as those underlying data are updated on the state's schedule.

All of these products are hosted or compatible with ArcGIS Online, so they can be shared among NCTA staff, chapter leaders, agency partners, and volunteers.

VI. Discussion and Recommendations

This project demonstrates how a relatively simple suitability model, when combined with thoughtfully prepared datasets, can provide valuable guidance for a complex trail-planning problem However, the results also highlight the importance of ground-truthing, partnership-building, and iterative modeling.

The most immediate recommendation is to continue and formalize the iterative workflow that volunteers have already been using. With the modeled route now available in KML format, NCTA can encourage field teams to download a segment, hike it, record their observations and preferred micro-reroutes, and then feed those edits back into the central geodatabase. Over time, these corrections and refinements can be used to adjust suitability scores, revisit the weight placed on certain features, and create a tighter fit between the model and the realities of the landscape.

In constrained areas, particularly near Ely and along major highways, strategic partnerships will be crucial. Utility corridors and transportation rights-of-way are sometimes the only places where continuous, legally defensible alignments can be achieved. Planners in Minnesota have access to a statewide right-of-way GIS resource maintained by MnDOT, which provides detailed maps of transportation right-of-way and related information across the state. This resource, available at https://www.dot.state.mn.us/maps/gisweb/row/, can help NCTA and partners identify where existing ROW may offer opportunities for co-located trail segments or parallel paths that remain within state or county control.

Another recommendation is to re-run the model in high-priority segments using higher resolution terrain data. LiDAR-derived bare-earth digital elevation models, slope maps, and flow-accumulation indices would allow the suitability surface to capture local landform patterns and drainage features that the 30-meter model cannot. In wet or steep areas, this could significantly change the relative attractiveness of alternative corridors. As higher-resolution hydrology and land cover datasets become available, they can be added to the same modeling framework using the existing pattern of classification, rasterization, and overlay.

It will also be valuable to experiment with alternative parcel-weighting scenarios. The current model treats all private parcels as uniformly difficult, but in practice some landowners are enthusiastic supporters of the trail, while others may be opposed or constrained by institutional requirements. Running multiple scenarios, such as one that strongly avoids all private land, one that assumes some private-land partnerships, and one that emphasizes minimizing road-walks even at the cost of more complex negotiations, would give NCTA a richer picture of potential tradeoffs. These scenarios could be shared with the NCTA board, agency partners, and funders to inform strategic decisions about where to invest staff and volunteer energy.

Finally, the workflow developed in this project can be extended to other NCT gaps in Minnesota and beyond. Because it relies primarily on publicly available datasets and documented processing steps, it offers a scalable framework for route planning across the entire trail system. As new data become available and as volunteers continue to walk and refine the route, the modeling framework can be updated and reused, ensuring that the tools remain relevant as the trail itself evolves.

VII. Conclusion

The work presented in this report represents a semester-long collaboration between the North Country Trail Association and graduate students at the University of Minnesota. By building a suitability model, generating candidate routes, and packaging results into accessible GIS products, North Star Consulting has provided a regional-scale view of where a future off-road NCT alignment between Grand Rapids and Snowbank Lake is most likely to succeed. The model

confirms the promise of areas already being explored by volunteers, such as corridors near the Taconite State Trail, while drawing attention to remaining bottlenecks in places like the Ely area.

At its core, the project is not about prescribing a single “correct” route, but about giving NCTA and its partners better tools for thinking about the landscape. The maps and datasets developed here can be used to focus ground-truthing where it will matter most, to organize outreach to landowners and ROW managers, and to support future environmental review and funding applications. As new data become available and as volunteers continue to walk and refine the route, the modeling framework can be updated and extended to other segments along the North Country Trail. In this way, the work of North Star Consulting forms one step in the long, collective effort to turn a line on a map into a continuous, safe, and inspiring trail on the ground.

VIII. Appendix

A. Data Sources

Data Layer	Source	Access
Current Proposed Trails	North Country Trail Association	
Trail Boundary	North Country Trail Association	
Parcels	State of MN	https://utility.arcgis.com/usrsvcs/servers/1627519e8d3f42bcb55532d48e9a61e5/rest/services/OpenParcels/plan_parcels_open/MapServer/0/0
Campgrounds	MN Geospatial commons	https://gisdata.mn.gov/dataset/struc-state-forest-campgrounds
Bridges	MN Geospatial commons	https://gisdata.mn.gov/dataset/trans-bridges
Current Trails	MN Geospatial commons	https://gisdata.mn.gov/dataset/trans-state-trails-minnesota
Bike Trails	MN Geospatial commons	https://gisdata.mn.gov/dataset/trans-bikeway
Snowmobile Trails	MN Geospatial Commons	https://gisdata.mn.gov/dataset/trans-snowmobile-trails-mn
Roads	MN Geospatial Commons	https://gisdata.mn.gov/dataset/trans-road-centerlines-open
ATV Trails	State of MN	https://gisdata.mn.gov/dataset/trans-state-park-trails-roads
Wetlands	Living Atlas	https://services.arcgis.com/P3ePLMYs2RVChkJx/arcgis/rest/services/USA_Wetlands/FeatureServer
Lakes	DNR - MN Geospatial Commons	https://gisdata.mn.gov/dataset/water-dnr-hydrography
Rivers	DNR - MN Geospatial Commons	https://gisdata.mn.gov/dataset/water-dnr-hydrography
Surface water	MN Geospatial Info Office	https://www.mngeo.state.mn.us/chouse/water.html

B. Billable Hours

Start of Week	End of Week	Total	Billable Hourly Rate (includes overhead and fringe costs)	Spending
10/28	11/3	36	$81.00	$2,916.00
11/4	11/10	51.5	$81.00	$4,171.50
11/11	11/17	79.25	$81.00	$6,419.25
11/18	11/24	57	$81.00	$4,617.00
11/25	12/1	37.1	$81.00	$3,005.10
12/2	12/8	65	$81.00	$4,941.00
		325.85		$26,069.85

C. Contact Information

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